Linear Algebra

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1 Vectors and Matrix Operations

1.1 Vectors

- **Vector** → a 1-dimensional array of numbers — has both magnitude and direction.
- **Linear Equation** → $Au = b$, where N is the number of training data points and M is the number of features.
- **Categorical (Non-numerical) Data** → encoded via **one-hot** vectors:

$$\text{red} = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \text{green} = \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \text{blue} = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

  One-hot vectors are very sparse. Any two distinct one-hot vectors are orthogonal, so they are never similar to each other.

- **Dot product and norm:**

$$\|\vec{v}\|^2 = v^T v, \qquad \|\vec{v}\| \to \text{Euclidean length (norm of } v).$$

- **Shadow (scalar projection)** → When $\|w\| = 1$, the scalar $v \cdot w = v^T w$ is the length of the orthogonal projection of v onto w (the "shadow" of v on w). In general: $\dfrac{v^T w}{\|w\|}$ is the scalar projection of v onto w.

1.2 Vector Operations

- Addition
- Subtraction
- Scalar multiplication
- **Inner product** (dot product) — notations all mean the same thing for real vectors:

$$a \cdot b \;=\; a^T b \;=\; b^T a \;=\; \langle a, b\rangle \;=\; \langle a \mid b\rangle \;=\; (a, b).$$

  Note: $a^T b = b^T a$ holds for real vectors; for complex vectors, $\langle a, b\rangle \neq \langle b, a\rangle$ in general.

1.3 Matrix–Vector Multiplication

- $A_{m\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$. Denoting the j-th column as $a_j \in \mathbb{R}^m$:

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

- **Column perspective** — Ax is a linear combination of the columns of A:

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n \in \mathbb{R}^m.$$

  Equivalently, the i-th entry of Ax is $a_i^T x$ (dot product of the i-th *row* of A with x).

- **Row perspective** — $x^T A$ (where $x \in \mathbb{R}^m$) is a linear combination of the *rows* of A. Denoting the i-th row as $\tilde{a}_i^T \in \mathbb{R}^{1\times n}$:

$$x^T A = x_1 \tilde{a}_1^T + x_2 \tilde{a}_2^T + \cdots + x_m \tilde{a}_m^T \in \mathbb{R}^{1\times n}.$$

- **Linear combination:** given vectors $a_1, \ldots, a_m$ and scalars $\beta_1, \ldots, \beta_m$, the result is $\displaystyle\sum_{i=1}^{m} \beta_i a_i$.

1.4 Matrix–Matrix Multiplication

Let $A \in \mathbb{R}^{m\times n}$ (columns $a_1,\dots,a_n$, rows $\tilde{a}_1^T,\dots,\tilde{a}_m^T$) and $B \in \mathbb{R}^{n\times p}$ (columns $b_1,\dots,b_p$, rows $\tilde{b}_1^T,\dots,\tilde{b}_n^T$). Then $C = AB \in \mathbb{R}^{m\times p}$:

- **Entry-wise**: $C_{ij} = \sum_{r=1}^{n} A_{ir}\,B_{rj}$ (dot product of i-th row of A with j-th column of B).
- **Column form** — each column of C is A times a column of B:
$$C = \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_p \end{bmatrix}.$$
- **Row form** — each row of C is a row of A times B:
$$C = \begin{bmatrix} \tilde{a}_1^T B \\ \tilde{a}_2^T B \\ \vdots \\ \tilde{a}_m^T B \end{bmatrix}.$$
- **Outer-product (rank-1) form**:
$$C = \sum_{r=1}^{n} a_r\,\tilde{b}_r^T = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \begin{bmatrix} \tilde{b}_1^T \\ \vdots \\ \tilde{b}_n^T \end{bmatrix}.$$

Algebraic properties. Matrix–vector multiplication is linear: $A(u+v) = Au + Av$, $(A+B)u = Au + Bu$, $(\alpha A)u = \alpha(Au) = A(\alpha u)$, $\mathbf{0}u = \mathbf{0}$, $A\mathbf{0} = \mathbf{0}$, and $Iu = u$. Matrix–matrix multiplication is **associative** ($(AB)C = A(BC)$) and **distributive** ($A(B+C) = AB + AC$), but *not commutative* ($AB \neq BA$ in general — the dimensions may not even match).

2 Systems of Linear Equations

2.1 Setup and Terminology

- A system of m equations in n unknowns:
$$A_{11}x_1 + \cdots + A_{1n}x_n = y_1, \quad \dots, \quad A_{m1}x_1 + \cdots + A_{mn}x_n = y_n.$$
Written compactly as $Ax = y$, where A is the $m\times n$ **coefficient matrix** and $[A \mid y]$ is the **augmented matrix**.
- $y = \mathbf{0} \to$ **homogeneous system**.
- **Equivalence**: Systems S_1 and S_2 are *equivalent* if every equation of each system is a linear combination of the equations of the other — they then have exactly the same solution set.

2.2 Gaussian Elimination and Elementary Matrices

- **Elementary Row Operations (ERO):**
  - **Scaling** — multiply one row by a nonzero scalar $c \neq 0$.
  - **Replacement** — add c times one row to another row.
  - **Interchange** — swap two rows.
- Each ERO is represented by left-multiplication by an **elementary matrix**. Examples (3×3):

Scaling (row 2): $\begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$, inverse: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{c} & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Replacement (row 1 += $c\cdot$ row 2): $\begin{bmatrix} 1 & c & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, inverse: $\begin{bmatrix} 1 & -c & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Interchange (rows 2&3): $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, inverse: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

- **Row-equivalence**: $A \xrightarrow{\text{EROs}} B$ means $B = E_k \cdots E_1 A = PA$, where $P = \prod E_i$ is a product of elementary matrices. Every ERO is invertible; its inverse is another ERO of the same type.
- Two matrices are **row-equivalent** iff one can be obtained from the other by a sequence of EROs, i.e., $A \sim B$. Row-equivalence is an **equivalence relation**: it is reflexive ($A \sim A$), symmetric ($A \sim B \Rightarrow B \sim A$), and transitive ($A \sim B$, $B \sim C \Rightarrow A \sim C$).
- **Theorem**: B is row-equivalent to $A \iff B = PA$ for some P that is a product of elementary matrices; and every elementary matrix is itself invertible.

2.3 Row Reduction and Pivots

- The **leading entry** (pivot candidate) of a nonzero row is its left-most nonzero entry.
- A column with a pivot → **basic variable**; a column without a pivot → **free variable**.
- **Solution types:**
  - No solution → **inconsistent**.
  - Exactly one solution → **consistent** (no free variables).
  - Infinitely many solutions → **consistent** (at least one free variable).

2.4 Echelon Form (EF)

A matrix is in **echelon form** if:

- All zero rows are at the bottom.
- Each leading entry is strictly to the right of the leading entry in the row above it.
- All entries *below* a leading entry (in the same column) are zero.

The pivots lie on or above the main diagonal, forming a staircase (upper-triangular shape).

2.5 Row-Reduced Echelon Form (RREF)

A matrix is in **RREF** if it is in echelon form and additionally:

- Every leading entry equals 1 (a *pivot of 1*).
- Each leading 1 is the *only* nonzero entry in its column (zeros above and below).

Theorem (Uniqueness): Every matrix $A_{m\times n}$ is row-equivalent to a *unique* RREF.

2.6 Consistency Criterion

A linear system $[A \mid b]$ is **consistent** iff the rightmost column of the augmented matrix is *not* a pivot column — equivalently, the echelon form contains no row of the form $[\,0 \;\cdots\; 0 \mid b\,]$ with $b \neq 0$.

2.7 Finding All Solutions

1. Write the augmented matrix $[A \mid b]$.
2. Reduce to EF and check consistency (if a zero row on the left has a nonzero b-entry, stop: inconsistent).
3. Reduce further to RREF.
4. Write the corresponding system; express each basic variable in terms of the free variables.
5. The general solution is: (particular solution) + (any element of $N(A)$).

2.8 Homogeneous Systems

- $Au = \mathbf{0}$ is always consistent: the **trivial solution** $x = \mathbf{0}$ always exists.
- **Non-trivial solutions** ($x \neq \mathbf{0}$) exist if and only if there is at least one free variable, i.e., $\text{rank}(A) < n$.

2.9 Existence of Solutions

$Au = b$ has a solution if and only if b is a linear combination of the columns of A, i.e., $b \in C(A)$.

3 LU Factorization and Complex Numbers

3.1 LU Factorization

- **Setting**: Gaussian elimination on square $A_{n\times n}$, applied to solve $Au = b$.
- **Without pivoting** (A can be reduced to echelon form without row interchanges):
$$A = LU \implies LUu = b.$$
Solve in two triangular steps: first $Ly = b$ (*forward substitution*), then $Ux = y$ (*backward substitution*).
- **With partial pivoting** (general case): a permutation matrix P is needed:
$$PA = LU \implies LUu = Pb.$$
- **How L is constructed**: if $E_k \cdots E_1 A = U$ (via elimination matrices), then $L = E_1^{-1} \cdots E_k^{-1}$.
- L is **unit lower triangular** (1's on diagonal); U is **upper triangular**.
- **Non-square**: $A_{m\times n} = L_{m\times m} U_{m\times n}$.
- **Existence**: A has an LU factorization (without pivoting) if and only if every leading principal submatrix of A is invertible (equivalently, A can be reduced to echelon form without row interchanges).

3.2 Complex Numbers and Algebraic Structures

- $\mathbb{C}$ is a plane; a complex number is $z = a + bi$ with $a, b \in \mathbb{R}$, $i^2 = -1$.
- **Conjugate**: $\bar{z} = \overline{a + bi} = a - bi$.
- **Modulus (magnitude)**: $|a + bi| = \sqrt{(a + bi)\overline{(a + bi)}} = \sqrt{a^2 + b^2}$.
- **Arithmetic**: $(a + bi) + (c + di) = (a + c) + (b + d)i$; $(a + bi)(c + di) = (ac - bd) + (bc + ad)i$; $\dfrac{a + bi}{c + di} = \dfrac{ac + bd}{c^2 + d^2} + \dfrac{bc - ad}{c^2 + d^2}\, i$.
- **Standard inner product over** $\mathbb{C}^n$ (conjugate-linear in first argument, linear in second — physics convention):
$$\langle x, y\rangle = \bar{x}_1 y_1 + \bar{x}_2 y_2 + \cdots + \bar{x}_n y_n = \bar{x}^T y.$$
- **Tuple** $\to$ an ordered list of numbers.
- **Binary operation** on a set A: any function $f : A \times A \to A$.
- **Closure**: a set is *closed* under an operation if applying the operation to any elements of the set always produces a result still within the set.

4 Fields and Vector Spaces

4.1 Fields

A **field** $(F, +, \cdot)$ is a set F with two binary operations satisfying:

- **Closure**: $a + b \in F$, $a \cdot b \in F$.
- **Associativity**: $a + (b + c) = (a + b) + c$; $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- **Commutativity**: $a + b = b + a$; $a \cdot b = b \cdot a$.
- **Identity elements**: 0 (additive: $a + 0 = a$), 1 (multiplicative: $a \cdot 1 = a$).
- **Inverses**: additive inverse $-a$; multiplicative inverse $\dfrac{1}{a}$ for $a \neq 0$.
- **Distributivity**: $a \cdot (b + c) = a \cdot b + a \cdot c$; $(a + b) \cdot c = a \cdot c + b \cdot c$.

Common fields: $\mathbb{R}$ (reals), $\mathbb{C}$ (complex numbers), $\mathbb{Q}$ (rationals).

4.2 Vector Spaces

A **vector space** $(V, F, +, \cdot)$ is a non-empty set V over a field F with:

- vector addition $+ : V \times V \to V$,
- scalar multiplication $\cdot : F \times V \to V$,

satisfying the 8 standard axioms (associativity, commutativity of addition, distributivity, etc.).

Key consequences:

- Every vector space has a *unique* additive identity $\mathbf{0}$.
- Every $v \in V$ has a *unique* additive inverse $-v$.
- $0 \cdot v = \mathbf{0}$ and $\alpha \cdot \mathbf{0} = \mathbf{0}$ for all $\alpha \in F$, $v \in V$.
- **Examples**: $\mathbb{R}^n$ over $\mathbb{R}$; $M_{m,n}(\mathbb{R})$; $P_n(\mathbb{R})$ (polynomials of degree $\leq n$, dimension $n+1$); $\mathbb{C}$ over $\mathbb{R}$. The set of **functions** $F^X = \{f : X \to F\}$ with $(f+g)(x) = f(x) + g(x)$ and $(af)(x) = af(x)$ is a vector space — as are its subsets of continuous / differentiable functions and all polynomials $P(\mathbb{R})$. Note $\mathbb{Z}$ over $\mathbb{R}$ is *not* a vector space (not closed under scalar multiplication).

4.3 Span and Subspaces

- **Coordinates**: any n-vector $b \in \mathbb{R}^n$ is a linear combination of the standard basis vectors $e_1, \ldots, e_n$: $b = \sum_{i=1}^{n} b_i e_i$.
- **Span**: the set of all linear combinations of a given set of vectors:

$$\text{Span}\{v_1, \ldots, v_k\} = \left\{ \sum_{i=1}^{k} c_i v_i \;\middle|\; c_i \in F \right\}.$$

- **Subspace**: a non-empty subset $W \subseteq V$ is a subspace iff it is closed under both vector addition and scalar multiplication — equivalently, for all $u, v \in W$ and $\alpha \in F$: $\alpha u + v \in W$. The zero vector $\mathbf{0}$ is always in a subspace.

5 Subspaces, Linear Independence, and Basis

5.1 Subspace Operations

- **Intersection**: If W_1, W_2 are subspaces of V, then $W_1 \cap W_2$ is a subspace — it is the *largest* subspace contained in both W_1 and W_2. More generally, the intersection of *any* collection of subspaces of V is a subspace.
- **Union**: $W_1 \cup W_2$ is a subspace if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$ (i.e., one contains the other).
- **Sum**: For subsets $A, B \subseteq V$, define $A + B = \{a + b \mid a \in A,\ b \in B\}$. If $W_1, \ldots, W_m$ are subspaces of V, then $W_1 + \cdots + W_m$ is also a subspace. In particular, $\text{Span}\{v_1, \ldots, v_p\}$ is a subspace of V.
- **Direct sum**: $U + W$ is a **direct sum** $U \oplus W$ if every vector in $U + W$ decomposes *uniquely* as $u + w$ with $u \in U$, $w \in W$.
  - Equivalently: U and W are subspaces of V with $U \cap W = \{\mathbf{0}\}$.

5.2 Linear Independence

- A list $\{v_1, \ldots, v_n\}$ is **linearly independent (LI)** if

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = \mathbf{0} \implies \lambda_1 = \cdots = \lambda_n = 0.$$

  Otherwise the list is **linearly dependent (LD)**.
- If $\mathbf{0} \in \{v_1, \ldots, v_n\}$, the list is linearly dependent.
- If a collection is LD, every superset is also LD.
- Every non-empty subset of an LI collection is LI.

- **Characterization of LD sets**: an indexed set $S = \{v_1, \ldots, v_n\}$ of two or more vectors is LD $\iff$ at least one vector is a linear combination of the others. In fact, if S is LD and $v_1 \neq \mathbf{0}$, then some v_j $(j > 1)$ is a linear combination of the *preceding* vectors $v_1, \ldots, v_{j-1}$. (Caution: this does *not* say every vector is a combination of the preceding ones.)
- If $v_1, \ldots, v_n$ are LD with $c_1 \neq 0$ in a dependence relation, then dropping v_1 does not shrink the span: $\text{Span}\{v_1, \ldots, v_n\} = \text{Span}\{v_2, \ldots, v_n\}$ (so spanning lists can be trimmed to a basis).
- If $v_1, \ldots, v_p$ are LI and $u = \beta_1 v_1 + \cdots + \beta_p v_p$, the coefficients β_i are *unique*.
- **Dimension bound** (key theorem): In a vector space of dimension d:
  - Any list of more than d vectors is linearly dependent.
  - Any LI list has *at most* d vectors; a list of exactly d LI vectors is a basis.
- **Algorithm for checking independence**:
  1. If the number of vectors $p > \dim(V)$: automatically dependent. Stop.
  2. If any vector is $\mathbf{0}$: dependent. Stop.
  3. Form $\lambda_1 v_1 + \cdots + \lambda_p v_p = \mathbf{0}$ and solve the system. If only solution is all $\lambda_i = 0$: independent. Otherwise: dependent.

5.3 Basis

A list $\{v_1, \ldots, v_n\}$ is a **basis** for V if it is:

- linearly independent, *and*
- spans V (every vector in V is a linear combination of $v_1, \ldots, v_n$).

5.4 Dimension

- A vector space is **finite-dimensional** if it is spanned by a finite set; otherwise **infinite-dimensional**.
- $\dim(V)$ = number of vectors in any basis of V (all bases of a finite-dimensional space have the same size — this is a theorem).
- $\dim(\{\mathbf{0}\}) = 0$ (the trivial vector space has dimension zero).
- Any set of $m > \dim(V)$ vectors is linearly dependent.
- Any set of $m < \dim(V)$ vectors cannot span V.

5.5 Basis of a Subspace and Dimension of Sums

- **Adjoining lemma**: if S is linearly independent and $u \notin \text{Span}(S)$, then $S \cup \{u\}$ is still linearly independent. (This is the engine behind extending an independent set to a basis.)
- If W is a subspace of a finite-dimensional V: every LI subset of W can be extended to a basis of W, and further to a basis of V.
- If W is a *proper* subspace of V $(W \neq V)$, then $\dim(W) < \dim(V)$.
- Every LI set in a finite-dimensional V is part of some basis.
- **Dimension of sum** (inclusion-exclusion):

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

  For three subspaces:

$$\begin{aligned}\dim(W_1 + W_2 + W_3) \leq\ & \dim(W_1) + \dim(W_2) + \dim(W_3) \\ & - \dim(W_1 \cap W_2) - \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) \\ & + \dim(W_1 \cap W_2 \cap W_3).\end{aligned}$$

  For *three* subspaces only the inequality holds — equality can fail (e.g. $W_1 = \text{Span}\{e_1\}$, $W_2 = \text{Span}\{e_2\}$, $W_3 = \text{Span}\{e_1 + e_2\}$ in $\mathbb{R}^2$).
- **Standard dimensions**: $\dim(F^n) = n$; $\dim(P_p) = p + 1$; $\dim(M_{m,n}) = mn$; and P (all polynomials), F (all functions), C (continuous functions) are all infinite-dimensional.

6 Coordinates, Spaces, and Rank

6.1 Dimensionality Theorems

Let V be a finite-dimensional vector space. Then:

- Any LI list in V can be *extended* to a basis of V.
- Any spanning list of V *contains* a basis as a sublist.
- Any LI list of length exactly $\dim(V)$ is automatically a basis.
- Any spanning list of length exactly $\dim(V)$ is automatically a basis.

6.2 Ordered Basis and Coordinate Vectors

- An **ordered basis** of V is a basis written as an ordered tuple $(b_1, \ldots, b_p)$.
- Given ordered basis $B = (b_1, \ldots, b_p)$ of V: for every $u \in V$ there exist *unique* scalars $c_1, \ldots, c_p$ with $u = c_1 b_1 + \cdots + c_p b_p$.

$$[u]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix} \quad \text{(coordinate vector of } u \text{ with respect to basis } B\text{)}.$$

6.3 Row Space and Column Space

- **Column space**: $C(A) = \{Au \mid u \in F^n\}$ — all linear combinations of the columns of A. This is a subspace of F^m.
- **Row space**: $R(A) = C(A^T)$ — all linear combinations of the rows of A. This is a subspace of F^n.
- **EROs and spaces**: Elementary row operations *do not change* $R(A)$, but they *do change* $C(A)$ as a set (though they preserve linear dependency relations among columns).
- **Row-equivalence**: $A \sim B \implies R(A) = R(B)$.
  - Non-zero rows of a row-echelon form of A form a basis for $R(A)$.
- **Pivot columns** of A (identified from the echelon form) form a basis for $C(A)$.

6.4 Null Space, Rank, Nullity, and the Four Fundamental Subspaces

- **Null space (kernel)**: $N(A) = \ker(A) = \{u \mid Au = \mathbf{0}\}$ — a subspace of F^n.
- **Left null space**: $N(A^T) = \{u \mid A^T u = \mathbf{0}\} = \{u \mid u^T A = \mathbf{0}\}$ — a subspace of F^m.
- **Rank**: $\text{rank}(A) = \dim(C(A)) = \dim(R(A))$ (column rank = row rank — a non-trivial theorem).
- **Nullity**: $\text{Nullity}(A) = \dim(N(A))$.
- **Rank-Nullity Theorem** for $A_{m\times n}$:

$$\text{rank}(A) + \text{Nullity}(A) = n.$$

- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
- **Column rank = row rank**: $\text{rank}(A) = \text{rank}(A^T)$. The proof uses two lemmas that are useful in their own right: $Ax = \mathbf{0} \iff A^T Ax = \mathbf{0}$ (so $N(A) = N(A^T A)$), hence $\text{rank}(A^T A) = \text{rank}(A)$; and $\text{rank}(BD) \leq \text{rank}(B)$.

Four Fundamental Subspaces of $A_{m\times n}$ with $\text{rank}(A) = r$:

$$\begin{array}{ll} C(A): & \text{subspace of } \mathbb{R}^m, \quad \dim = r. \\ R(A) = C(A^T): & \text{subspace of } \mathbb{R}^n, \quad \dim = r. \\ N(A): & \text{subspace of } \mathbb{R}^n, \quad \dim = n - r. \\ N(A^T): & \text{subspace of } \mathbb{R}^m, \quad \dim = m - r. \end{array}$$

Moreover, $C(A) \perp N(A^T)$ and $R(A) \perp N(A)$ (orthogonal complements in their respective spaces).

Solvability of $Ax = b$ by shape (rank r):

shape	$\dim R(A), C(A)$	$\dim N(A), N(A^T)$	$Ax = b$
$m = n = r$	$r,\ r$	$0,\ 0$	unique $x = A^{-1}b$
$m > n = r$ (tall)	$r,\ r$	$0,\ m - r$	solvable iff $b \in C(A)$; unique
$n > m = r$ (wide)	$r,\ r$	$n - r,\ 0$	always solvable; $x = x_p + N(A)$
$r < \min(m, n)$	$r,\ r$	$n - r,\ m - r$	solvable iff $b \in C(A)$; ∞ solutions

7 Linear Transformations

7.1 Definition and Basic Properties

- A map $T : V \to W$ is a **linear transformation** (linear map) if for all $u, v \in V$ and $\alpha \in F$:

$$T(\alpha u + v) = \alpha T(u) + T(v). \quad \text{(linearity / superposition)}$$

  Equivalently: $T(\mathbf{0}) = \mathbf{0}$ and $T\big(\sum_i \alpha_i v_i\big) = \sum_i \alpha_i T(v_i)$.

- For a matrix $A \in F^{m\times n}$, $T_A(u) = Au$ is linear with domain F^n and codomain F^m.
- $\mathcal{L}(V, W) \to$ the set of all linear maps from V to W; this is itself a vector space.
- $\mathcal{L}(V) := \mathcal{L}(V, V) \to$ linear operators on V.
- **Unique determination**: given an ordered basis $(v_1, \ldots, v_n)$ of V and any list $(w_1, \ldots, w_n)$ in W, there is a *unique* $T \in \mathcal{L}(V, W)$ with $T(v_i) = w_i$ for all i.
- **Composition**: If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then $S \circ T \in \mathcal{L}(U, W)$, written ST, with $ST(u) = S(T(u))$.

7.2 Standard Transformations in $\mathbb{R}^2$

- **Rotation** by angle θ (counterclockwise):

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

- **Projection** onto the line through the origin at angle θ:

$$P_\theta = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}, \qquad P_\theta^2 = P_\theta \quad \text{(idempotent)}.$$

- **Reflection** across the line through the origin at angle θ (use Ref_θ to avoid clash with rotation):

$$\mathrm{Ref}_\theta = \begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}.$$

  Note: $\mathrm{Ref}_\theta^2 = I$ (reflecting twice returns to original).

- **Norms** are *not* linear maps — they fail $\|cv\| = c\|v\|$ for negative scalars ($c < 0$ gives $|c|\|v\| \neq c\|v\|$), and the triangle inequality $\|u + v\| \leq \|u\| + \|v\|$ replaces additivity.
- **Translation** by a fixed nonzero vector t is *not* a linear map — it does not satisfy $T(\mathbf{0}) = \mathbf{0}$ (since $\mathbf{0}+t = t \neq \mathbf{0}$). It is an **affine** map.
- **Other common maps**: uniform scaling $S = sI$; non-uniform scaling $\mathrm{diag}(s_x, s_y)$; shear $\begin{bmatrix} 1 & \lambda \\ 0 & 1 \end{bmatrix}$ (and its transpose). Powers of a rotation compose the angle: $R_\theta^n = R_{n\theta}$, i.e. $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^n = \begin{bmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{bmatrix}$. **Selector** matrices (rows are standard unit vectors) pick out / reorder / down-sample entries of x.

7.3 Null Space, Range, and the Fundamental Theorem

- **Null space**: $\text{Null}(T) = \{v \in V \mid T(v) = \mathbf{0}\}$ — a subspace of V; $\text{Nullity}(T) = \dim(\text{Null}(T))$.
- **Range**: $\text{range}(T) = \{T(v) \mid v \in V\}$ — a subspace of W.
- **Fundamental Theorem of Linear Maps** (Rank-Nullity for maps):

$$\dim(V) = \text{Nullity}(T) + \dim(\text{range}(T)).$$

- **Corollaries**: a linear map to a *lower*-dimensional space ($\dim W < \dim V$) is never injective; a linear map to a *higher*-dimensional space ($\dim W > \dim V$) is never surjective.

7.4 Injective, Surjective, Bijective

- **Injective (one-to-one)**: $T(v_1) = T(v_2) \Longrightarrow v_1 = v_2$.
  - T is injective $\iff$ $\text{Null}(T) = \{\mathbf{0}\}$ $\iff$ $T(u) = \mathbf{0}$ has only the trivial solution.
  - For matrix A: T_A injective $\iff$ columns of A are linearly independent $\iff$ $\text{rank}(A) = n$.
  - $T(u) = b$ has *at most* one solution.
  - Wide matrices ($A_{m\times n}$, $n > m$) cannot be injective (more columns than rows means columns must be dependent).
- **Surjective (onto)**: For every $w \in W$, $T(v) = w$ has at least one solution, i.e., $\text{range}(T) = W$.
  - For matrix A: T_A surjective $\iff$ $\text{rank}(A) = m$ (rows are linearly independent).
  - Tall matrices ($A_{m\times n}$, $m > n$) cannot be surjective (rank $\leq n < m$).
- **Bijective**: injective *and* surjective.
  - For matrix $A_{m\times n}$: T_A bijective $\iff$ $m = n$ and A has full rank (i.e., A is a square invertible matrix).
  - $T(u) = b$ has *exactly* one solution.
- **Square case**: for a *square* matrix, injective $\iff$ surjective (a pivot in every column $\iff$ a pivot in every row). So a self-map $T : \mathbb{R}^n \to \mathbb{R}^n$ is one-to-one iff it is onto. Conversely, if a matrix map is both injective and surjective then $m = n$.

8 Invertibility, Isomorphisms, and Matrix Inverse

8.1 Invertible Linear Maps and Isomorphisms

- $T \in \mathcal{L}(V, W)$ is **invertible** if there exists $S \in \mathcal{L}(W, V)$ such that $ST = I_V$ (identity on V) and $TS = I_W$ (identity on W). Then $S = T^{-1}$ is the *unique* inverse.
- T invertible $\iff$ T bijective.
- An invertible linear map is an **isomorphism**; V and W are **isomorphic** ($V \cong W$) if an isomorphism between them exists.
- $T \in \mathcal{L}(V, W)$ bijective $\Rightarrow V \cong W$.
- If $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, U)$ are both bijective, then $V \cong W \cong U$, so $V \cong U$ over F.
- **Theorem**: Two finite-dimensional vector spaces over F are isomorphic $\iff$ they have the same dimension.

8.2 Change of Basis

- Given two ordered bases $B = (b_1, \ldots, b_n)$ and $C = (c_1, \ldots, c_n)$ of V:

$$[u]_C = P_{C\leftarrow B}\,[u]_B.$$

- **Change-of-basis matrix**: $P_{C\leftarrow B} = \big[[b_1]_C \ \cdots \ [b_n]_C\big]$ (columns are coordinates of b_i in basis C).
- $P_{C\leftarrow B}$ is always invertible, and $(P_{C\leftarrow B})^{-1} = P_{B\leftarrow C}$.

8.3 Matrix Representation of a Linear Map

Let $T \in \mathcal{L}(V, W)$, $E = (e_1, \ldots, e_n)$ an ordered basis of V, $B = (b_1, \ldots, b_m)$ an ordered basis of W. The **matrix of** T with respect to E and B is the $m \times n$ matrix:

$$\mathcal{M}(T)_{E,B} = \big[[T(e_1)]_B \ \cdots \ [T(e_n)]_B\big], \quad \text{so that} \quad [T(u)]_B = \mathcal{M}(T)_{E,B}\,[u]_E.$$

8.4 Matrix Inverse

- For square $A \in M_{n\times n}$: B is the **inverse** of A if $AB = BA = I_n$.
  - If no such B exists, A is **singular** (non-invertible).
  - The inverse is *unique* (if it exists); written A^{-1}.
  - A invertible $\iff$ columns of A are linearly independent $\iff$ $\operatorname{rank}(A) = n$ $\iff$ RREF of A is I_n.
- **Left inverse**: X satisfying $XA = I$ is a left inverse of A; A is **left-invertible** if such X exists.
  - A left-invertible $\iff$ columns of A are linearly independent (i.e., A has full column rank).
  - $Au = b \xrightarrow{X \text{ is left inv.}} u = Xb$ gives the unique solution.
  - Wide matrices ($n > m$) are not left-invertible.
  - The left inverse is generally *not* unique for non-square matrices.
- **Left inverse of a vector** $a \in \mathbb{R}^n$: since $a^T a = \|a\|^2$, we get $\dfrac{a^T a}{\|a\|^2} = 1$, so the pseudo-inverse of a (as a column matrix) is:

$$a^{-1} = \frac{a^T}{\|a\|^2} \quad \text{(a } 1 \times n \text{ left inverse of the } n \times 1 \text{ vector } a\text{)}.$$

- **Right inverse**: X satisfying $AX = I$ is a right inverse of A; A is **right-invertible** if such X exists.
  - A right-invertible $\iff$ rows of A are linearly independent (full row rank).
  - $Au = b \Rightarrow u = Xb$ is *a* solution for every b (not necessarily unique).
  - Tall matrices ($m > n$) are not right-invertible.
- A is right-invertible $\iff$ A^T is left-invertible (transpose $AX = I$ to get $X^T A^T = I$).
- **Square matrices**: left inverse = right inverse = A^{-1} (and rows and columns are both linearly independent).
- 2×2 **formula**: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, valid iff $ad - bc \neq 0$.

8.5 Gauss-Jordan Elimination and Inverse Properties

- **Gauss-Jordan method** to find A^{-1}: form augmented $[A \mid I_n]$, reduce to RREF:
  - RREF $\to [I_n \mid A^{-1}]$ ✓
  - RREF $\to [B \mid C]$ with $B \neq I_n$: A is singular (not invertible).
- **Inverse identities**:
  - $(A^{-1})^{-1} = A$
  - $(A^k)^{-1} = (A^{-1})^k$
  - $(cA)^{-1} = \dfrac{1}{c}\,A^{-1} \quad (c \neq 0)$
  - $(A^T)^{-1} = (A^{-1})^T$
  - $(AB)^{-1} = B^{-1}A^{-1}$ (reverse order!)
- **Solution set**: the complete solution of $Au = b$ is $K = \{s_p + n \mid n \in N(A)\}$, where s_p is any particular solution.

8.6 Triangular Matrices and Rank

- Upper or lower triangular matrices with all nonzero diagonal entries are invertible.
- The change-of-basis matrix is always invertible.
- $\text{rank}(A) = \text{rank}(A^{-1})$ for invertible A.
- If $B_{n\times n}$ is invertible: $\text{rank}(AB) = \text{rank}(A)$ (multiplying by invertible matrix preserves rank).

8.7 The Invertible Matrix Theorem

For a square $A \in M_{n\times n}$, the following are **all equivalent** — if one holds, they all do:

- A is invertible (nonsingular); A^T is invertible.
- A is row-equivalent to I_n; the RREF of A is I_n; A has n pivot positions.
- $\det A \neq 0$; 0 is not an eigenvalue of A.
- The columns of A are linearly independent and span $\mathbb{R}^n$; likewise the rows. $\text{rank}(A) = n$.
- $Ax = \mathbf{0}$ has only the trivial solution; $Ax = b$ has exactly one solution for every b.
- The map $x \mapsto Ax$ is one-to-one and onto ($\mathbb{R}^n \to \mathbb{R}^n$).
- A has a left inverse ($\exists\, C : CA = I$) and a right inverse ($\exists\, D : AD = I$); these coincide with A^{-1}.

9 Linear Functionals and Bilinear Forms

9.1 Linear Functionals

- A **linear functional** $f : V \to F$ is a linear map into the scalar field. For $V = \mathbb{R}^n$, f satisfies the **superposition property**:
$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y).$$
- **Riesz representation**: every linear functional on $\mathbb{R}^n$ has the unique representation $f(u) = a^T u$ for some fixed $a \in \mathbb{R}^n$. Equivalently, every linear functional is an inner product with a fixed vector.
- **Dual space**: $V^* = \mathcal{L}(V, F)$, the set of all linear functionals on V. V^* is itself a vector space with $\dim(V^*) = \dim(V)$.

9.2 Bilinear Forms

- A **bilinear form** is a map $f : V \times W \to F$ that is linear in *both* arguments separately.
- A **symmetric bilinear form** on V satisfies $f(v, w) = f(w, v)$ for all $v, w \in V$. With respect to a basis, it is represented by a *symmetric* matrix $A = A^T$:
$$f(v, w) = v^T A w = w^T A v.$$
- **Associated (Gram) matrix**: given ordered basis $B = (b_1, \ldots, b_n)$ of V and bilinear form $f : V \times V \to F$, the matrix $[f]_B \in F^{n\times n}$ has entries
$$([f]_B)_{ij} = f(b_i, b_j), \quad \text{so that} \quad f(v, w) = [v]_B^T\, [f]_B\, [w]_B.$$
- **Sesquilinear form over** $\mathbb{C}$: a map $f : V \times V \to \mathbb{C}$ that is linear in the first argument and *conjugate-linear* (anti-linear) in the second:
$$f(v, \lambda w) = \bar{\lambda}\, f(v, w).$$
This is the correct generalization of bilinear forms to complex spaces, needed for Hermitian inner products.

10 Inner Product Spaces

10.1 Inner Product Axioms

An **inner product** on a vector space V over F ($= \mathbb{R}$ or $\mathbb{C}$) is a map $\langle \cdot, \cdot \rangle : V \times V \to F$ satisfying, for all $u, v, w \in V$ and $c \in F$:

- **Positive definiteness**: $\langle v, v \rangle \geq 0$; and $\langle v, v \rangle = 0 \iff v = \mathbf{0}$.
- **Conjugate symmetry**: $\langle w, v \rangle = \overline{\langle v, w \rangle}$ (reduces to $\langle w, v \rangle = \langle v, w \rangle$ over $\mathbb{R}$).
- **Linearity in first argument**: $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$; $\langle cv, w \rangle = c\langle v, w \rangle$.

Note: conjugate linearity in the second argument follows: $\langle v, cw \rangle = \bar{c}\langle v, w \rangle$. Over $\mathbb{R}$, an inner product is exactly a **positive-definite symmetric bilinear form**, so the four real axioms (symmetry, additivity, homogeneity in the first slot, positive-definiteness) suffice — the second-slot properties follow. Also $\langle \mathbf{0}, v \rangle = \langle v, \mathbf{0} \rangle = 0$.

Standard examples:

$$\langle x, y \rangle = x^T y \quad (\text{over } \mathbb{R}); \qquad \langle x, y \rangle = \bar{x}^T y = \sum_{i=1}^{n} \bar{x}_i y_i \quad (\text{over } \mathbb{C}).$$

10.2 Inner Products for Matrices and Functions

- **Frobenius inner product** for matrices ($X, Y \in \mathbb{R}^{m \times n}$):

$$\langle X, Y \rangle = \operatorname{trace}(X^T Y) = \sum_i \sum_j X_{ij} Y_{ij}.$$

- L^2 **inner product** for functions: $C[a, b]$ = continuous real functions on $[a, b]$; for $f, g \in C[a, b]$:

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt.$$

- An **inner product space** (real: Euclidean space; complex: Unitary/Hilbert space) is a vector space equipped with an inner product.

10.3 Norms and Metrics

- Every inner product induces a **norm**: $\|v\| = \sqrt{\langle v, v \rangle}$.
- $\|v\|$ is the **length** of v; $\|u - v\|$ is the **distance** between u and v (a metric).
- $\|v\| = 1 \Rightarrow v$ is a **unit vector**. **Normalizing**: $\hat{v} = \dfrac{v}{\|v\|}$ is a unit vector.
- **Norm axioms**: a norm $\| \cdot \| : V \to \mathbb{R}_{\geq 0}$ satisfies:
  - $\|\alpha v\| = |\alpha|\, \|v\|$ **(absolute homogeneity)** — note the *absolute value* around α.
  - $\|u + v\| \leq \|u\| + \|v\|$ **(triangle inequality)**.
  - $\|v\| = 0 \iff v = \mathbf{0}$ **(non-degeneracy)**.
- **Norm expansion**:

$$\|u + v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v \rangle + \|v\|^2.$$

Over $\mathbb{R}$, $\mathrm{Re}\langle u, v \rangle = \langle u, v \rangle = u^T v$, so $\|u + v\|^2 = \|u\|^2 + 2u^T v + \|v\|^2$.

11 Orthogonality and Gram-Schmidt

11.1 Orthogonal and Orthonormal Sets

- Vectors u, v are **orthogonal** ($u \perp v$) if $\langle u, v\rangle = 0$.
- A set $\{a_1, \ldots, a_k\}$ is **orthogonal** if $\langle a_i, a_j\rangle = 0$ for all $i \neq j$.
- It is **orthonormal** if additionally $\|a_i\| = 1$ for all i:

$$\langle a_i, a_j\rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

- **Pythagorean theorem**: $u \perp v \iff \|u+v\|^2 = \|u\|^2 + \|v\|^2$.
- **Theorem**: An orthogonal set of *nonzero* vectors is linearly independent, and forms a basis for its span.
- If $\{a_1, \ldots, a_n\}$ is orthonormal in $\mathbb{R}^n$, it is an orthonormal basis for $\mathbb{R}^n$.
- **Expansion (Fourier) coefficients**: if $\{a_1, \ldots, a_k\}$ is orthonormal, then $y \in \text{Span}\{a_i\}$ has the direct expansion

$$y = (a_1^T y)\, a_1 + \cdots + (a_k^T y)\, a_k.$$

  For a merely *orthogonal* set, $y = \sum_j c_j a_j$ with $c_j = \dfrac{y \cdot a_j}{a_j \cdot a_j}$ — no linear system to solve.

11.2 Orthogonal Subspaces and Complement

- Subspaces $W_1, W_2 \subseteq V$ are **orthogonal** ($W_1 \perp W_2$) if $\langle w_1, w_2\rangle = 0$ for all $w_1 \in W_1$, $w_2 \in W_2$.
  - Equivalent: the bases of W_1 and W_2 are pairwise orthogonal.
- **Orthogonal complement** of $W \subseteq V$:

$$W^\perp = \{v \in V \mid \langle v, w\rangle = 0 \text{ for all } w \in W\}.$$

  $W^\perp$ is a subspace of V, and $W^\perp \cap W = \{\mathbf{0}\}$.
- Orthogonal subspaces need not be complements: $W_1 = \text{Span}(e_1)$ and $W_2 = \text{Span}(e_2)$ in $\mathbb{R}^3$ are orthogonal but do not span $\mathbb{R}^3$.
- In a finite-dimensional inner product space: $V = W \oplus W^\perp$, so $\dim(W^\perp) = \dim(V) - \dim(W)$.
- **Fundamental orthogonality**: $C(A) \perp N(A^T)$ and $R(A) \perp N(A)$.

11.3 Gram-Schmidt Orthogonalization

Given a list $S = (a_1, \ldots, a_k)$ in an inner product space, the **Gram-Schmidt** process constructs an orthonormal list $(q_1, \ldots, q_r)$ that is a basis for $\text{Span}(S)$:

- Step 1: $\tilde{q}_1 = a_1, \quad q_1 = \dfrac{\tilde{q}_1}{\|\tilde{q}_1\|}$.
- Step 2: $\tilde{q}_2 = a_2 - \langle q_1, a_2\rangle\, q_1, \quad q_2 = \dfrac{\tilde{q}_2}{\|\tilde{q}_2\|}$.
- Step 3: $\tilde{q}_3 = a_3 - \langle q_1, a_3\rangle\, q_1 - \langle q_2, a_3\rangle\, q_2, \quad q_3 = \dfrac{\tilde{q}_3}{\|\tilde{q}_3\|}$.
- **General step** i:

$$\tilde{q}_i = a_i - \sum_{j=1}^{i-1} \langle q_j,\, a_i\rangle\, q_j, \qquad q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}.$$

- **If** $\tilde{q}_i = \mathbf{0}$: skip (do not produce q_i). This happens iff $a_i \in \text{Span}\{a_1, \ldots, a_{i-1}\}$, i.e., S is linearly dependent and:

$$a_i = \langle q_1, a_i\rangle q_1 + \cdots + \langle q_{i-1}, a_i\rangle q_{i-1}.$$

- If no step is skipped $\Rightarrow$ S is linearly independent and $(q_1, \ldots, q_k)$ is an orthonormal basis for $\text{Span}(S)$.
- **Complexity**: $O(nk^2)$ (where n is the dimension of each vector).

Theorem (Existence of Orthonormal Basis): Every finite-dimensional inner product space has an orthonormal basis (apply Gram-Schmidt to any basis).

12 Determinants

12.1 Geometric Meaning

- $|\det A|$ measures how much A scales space: it is the area (2D), volume (3D), or hypervolume of the image of the unit square/cube under the linear map A.
- A 2×2 matrix A maps the unit square (sides e_1, e_2) to a parallelogram with sides Ae_1, Ae_2 (the columns of A); $\det A$ is the signed area of that parallelogram.
- For T with matrix A and a region S: $\text{area}(T(S)) = |\det A| \cdot \text{area}(S)$ (and analogously for volume in $\mathbb{R}^3$).
- A rotation $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ has determinant 1 (preserves area).

12.2 Volume as an Alternating Multilinear Form

The volume $V(a_1, \ldots, a_n)$ of the parallelepiped with legs $a_1, \ldots, a_n$ is characterized by:

- **Multilinearity (scaling)**: $V(\ldots, \lambda a_i, \ldots) = \lambda V(\ldots, a_i, \ldots)$.
- **Multilinearity (additivity)**: $V(\ldots, a_i + \omega, \ldots) = V(\ldots, a_i, \ldots) + V(\ldots, \omega, \ldots)$.
- **Alternating**: swapping two legs flips the sign, $V(a_1, a_2) = -V(a_2, a_1)$; hence $V = 0$ whenever two legs are equal (or linearly dependent).
- **Normalization**: $V(e_1, \ldots, e_n) = 1$.

Expanding $V\left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix}\right)$ by multilinearity gives $ad - bc$ — the familiar 2×2 formula.

12.3 Multilinear and Alternating Forms

- A **multilinear form** $f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_p \to F$ is linear in each argument separately.
- An m-linear form α on V is **alternating** if $\alpha(v_1, \ldots, v_m) = 0$ whenever two of the inputs are equal. Equivalently (bilinear case) $\alpha(u, w) = -\alpha(w, u)$.
- $V^2 = V^2_{\text{sym}} \oplus V^2_{\text{alt}}$; and V^m_{alt} is a subspace of V^m.
- **Linear dependence**: if $v_1, \ldots, v_m$ is linearly dependent, then $\alpha(v_1, \ldots, v_m) = 0$ for every alternating α. There is no nonzero alternating m-linear form when $m > \dim V$.
- **Swaps**: for alternating α, each transposition of two inputs multiplies the value by -1.

12.4 Permutations and the Leibniz Formula

- A **permutation** of $(1, \ldots, m)$ is a reordering $(j_1, \ldots, j_m)$; the set is $\text{perm}(m)$.
- The **sign** is $\text{sign}(j_1, \ldots, j_m) = (-1)^N$, where N counts inversions (pairs $k < l$ with k appearing after l). The identity permutation has sign $+1$.
- For an alternating form, $\alpha(v_{j_1}, \ldots, v_{j_m}) = \text{sign}(j_1, \ldots, j_m)\, \alpha(v_1, \ldots, v_m)$.
- **Leibniz formula**: for an $n \times n$ matrix A,

$$\det A = \sum_{(j_1, \ldots, j_n) \in \text{perm}(n)} \text{sign}(j_1, \ldots, j_n)\, A_{j_1,1}\, A_{j_2,2} \cdots A_{j_n,n}.$$

The determinant is the unique alternating n-linear form (as a function of the columns) with $\det I = 1$.

12.5 Cofactor Expansion

- Let A_{ij} be the submatrix obtained by deleting row i and column j. The (i,j)**-cofactor** is $C_{ij} = (-1)^{i+j} \det(A_{ij})$.
- **Recursive definition** (expansion along the first row):

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j}) = \sum_{j=1}^{n} a_{1j} C_{1j}.$$

- The expansion may be taken along *any* row i or *any* column j:

$$\det A = \sum_{k=1}^{n} a_{ik} C_{ik} = \sum_{k=1}^{n} a_{kj} C_{kj}.$$

- **Closed forms**:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc, \qquad \det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - afh - bdi - ceg.$$

- **Adjugate**: $\operatorname{adj}(A) = C^T$ (transpose of the cofactor matrix), giving

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

12.6 Properties of the Determinant

- **Row operations**: adding a multiple of one row to another leaves det unchanged; swapping two rows flips the sign; scaling a row by k multiplies det by k. Hence $\det(kA_{n\times n}) = k^n \det A$.
- A zero row/column, two equal rows/columns, or linearly dependent columns all force $\det A = 0$. In fact columns are linearly dependent $\iff \det A = 0$.
- **Triangular**: det of a (upper/lower) triangular matrix is the product of its diagonal entries; $\det I = 1$.
- **Multiplicativity**: $\det(AB) = \det(A)\det(B)$; but det is *not* linear ($\det(A+B) \neq \det A + \det B$ in general).
- **Invertibility**: A is invertible $\iff \det A \neq 0$, and then $\det(A^{-1}) = 1/\det A$.
- **Transpose**: $\det(A^T) = \det A$. For orthogonal/unitary U: $|\det U| = 1$.

12.7 Cramer's Rule

Let A be invertible and $A_i(b)$ denote A with its i-th column replaced by b. The unique solution of $Ax = b$ has entries

$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, \ldots, n.$$

Applying this to $Ax = e_j$ recovers the adjugate formula $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$.

13 Eigenvalues and Eigenvectors

13.1 Definitions

- For a square A, a nonzero v with $Av = \lambda v$ is an **eigenvector** and λ the corresponding **eigenvalue**. An eigenvector must be nonzero, but $\lambda = 0$ is allowed.
- **Eigenspace**: $E_\lambda = N(A - \lambda I) = \operatorname{Span}\{\text{eigenvectors for } \lambda\} \cup \{\mathbf{0}\}$, a subspace of $\mathbb{R}^n$.
- **Spectrum**: $\sigma(A)$ is the set of all eigenvalues of A.

13.2 Characteristic Polynomial

- $Av = \lambda v \iff (A - \lambda I)v = \mathbf{0}$ with $v \neq \mathbf{0}$, so $A - \lambda I$ must be *singular*: $\det(A - \lambda I) = 0$.
- **Characteristic polynomial** $p(\lambda) = \det(A - \lambda I)$; **characteristic equation** $p(\lambda) = 0$. To get a monic polynomial (leading coefficient $+1$) one often uses $\det(\lambda I - A) = (-1)^n \det(A - \lambda I)$.
- An $n \times n$ matrix has n eigenvalues counted with multiplicity (over $\mathbb{C}$).
- Writing $p(\lambda) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$: the coefficient $c_{n-1} = -\text{trace}(A)$ and $c_0 = (-1)^n \det A$.

13.3 Key Facts

- **Sum** of eigenvalues $= \text{trace}(A)$; **product** of eigenvalues $= \det(A)$.
- $0 \in \sigma(A) \iff \det A = 0 \iff A$ singular. Thus A is invertible $\iff 0$ is not an eigenvalue.
- **Triangular matrices**: the eigenvalues are the diagonal entries.
- **Structured matrices**: a projection has eigenvalues $\{0, 1\}$ (with 0 of multiplicity $n - r$ and 1 of multiplicity $r = \text{rank}$); reflection and permutation matrices have eigenvalues among $\{1, -1\}$.
- The nonzero eigenvalues of AB equal the nonzero eigenvalues of BA.

14 Diagonalization

14.1 Similarity

- A and B are **similar** if $A = Q^{-1}BQ$ for some invertible Q. This is exactly a change of basis: if $\bar{A} = P^{-1}AP$, then $\bar{A}$ is the matrix of the same linear map in the new basis.
- **Similarity invariants**: similar matrices share determinant, characteristic polynomial, eigenvalues, trace, and rank; their inverses (when they exist) are also similar.
- **Warnings**: equal eigenvalues do *not* imply similarity — e.g. $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is not similar to $2I$. Similarity is *not* row-equivalence; row operations generally change eigenvalues.

14.2 Diagonalizable Matrices

- A is **diagonalizable** if $A = Q\Lambda Q^{-1}$ with Λ diagonal. Stacking $Aq_i = \lambda_i q_i$ into columns gives $AQ = Q\Lambda$, so the columns of Q are eigenvectors and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$.
- $A_{n\times n}$ is diagonalizable $\iff A$ has n linearly independent eigenvectors. **Corollary**: n distinct eigenvalues $\Rightarrow$ diagonalizable.
- **Multiplicities**: the *algebraic* multiplicity of λ is its multiplicity as a root of $p(\lambda)$; the *geometric* multiplicity is $\dim E_\lambda$. Always geometric $\leq$ algebraic.
- A is diagonalizable $\iff$ the geometric multiplicities sum to $n \iff$ for every eigenvalue geometric $=$ algebraic. E.g. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is *not* diagonalizable.

14.3 Powers and Cayley–Hamilton

- Diagonalization turns matrix powers into scalar powers: $A^k = Q\Lambda^k Q^{-1}$, with $\Lambda^k = \text{diag}(\lambda_1^k, \ldots, \lambda_n^k)$.
- **Cayley–Hamilton**: every square matrix satisfies its own characteristic polynomial, $p(A) = 0$. This reduces high powers of A to low ones, lets A^{-1} be written as a polynomial in A (when invertible), and computes powers even of non-diagonalizable matrices.

15 Matrix Factorizations

15.1 QR Factorization

- If $A_{m\times n}$ has linearly independent columns, then $A = QR$ where Q has orthonormal columns ($Q^TQ = I$) and R is upper triangular with positive diagonal.
- Q comes from running Gram-Schmidt on the columns of A; then $R = Q^TA$, with $R_{jk} = \langle a_k, q_j\rangle$ for $j < k$ and $R_{ii} = \|\tilde{q}_i\|$.
- If A is square, Q is orthogonal ($Q^TQ = QQ^T = I$). Every matrix has a QR factorization, though R need not be invertible (for rank-deficient A, R is block upper-triangular).
- **Determinant**: if $A = UT$ with U unitary and T upper triangular, then $|\det A| = |t_{11}t_{22}\cdots t_{nn}|$.

15.2 Schur Triangularization

For every $A \in M_n(\mathbb{C})$ there exist a unitary U and an upper-triangular T with

$$A = UTU^*.$$

The diagonal of T holds the eigenvalues of A. Schur forms are highly non-unique. A real matrix with non-real eigenvalues (e.g. $\begin{bmatrix}1 & -2\\ 1 & -1\end{bmatrix}$) has no real Schur triangularization, only a complex one.

15.3 Spectral Decomposition

- **Complex**: $A = UDU^*$ with U unitary and D diagonal $\iff$ A is **normal** ($A^*A = AA^*$).
- **Real**: $A = UDU^T$ with U orthogonal and D real diagonal $\iff$ A is **symmetric** ($A = A^T$).
- Eigenvectors of a normal matrix belonging to distinct eigenvalues are orthogonal.
- Spectral decomposition applies only to normal (complex) / symmetric (real) matrices; the general tool for *all* matrices is the SVD.

15.4 LU and PLU Factorization

- $A = LU$ with L unit lower-triangular and U upper-triangular. Construct U by Gaussian elimination (no row swaps); place the reciprocals/negatives of the elimination multipliers into L.
- To solve $Ax = b$: set $Ux = y$, solve $Ly = b$ (forward substitution), then $Ux = y$ (back substitution).
- **Existence**: an invertible A has an LU factorization $\iff$ all leading principal minors are nonzero; a rank-k singular matrix has one if its first k leading principal minors are nonzero.
- **PLU** (partial pivoting): any nonsingular A factors as $A = PLU$ with P a permutation matrix; equivalently factor $P^TA = LU$. Also called Gaussian elimination with partial pivoting (GEPP); not unique.
- **Cost**: LU takes $\approx \frac{2}{3}n^3$ flops (vs. $\approx 2n^3$ to form A^{-1}); each triangular solve is $\approx 2n^2$ flops; sparsity of A tends to carry over to L, U but not to A^{-1}.

15.5 Cholesky and Low-Rank Factorization

- **Cholesky**: every positive definite A factors as $A = R^TR$ with R upper-triangular and positive diagonal (R is the "square root" of A). Cost $\approx \frac{1}{3}n^3$ flops; it also serves as a practical positive-definiteness test.
- **Low-rank**: a rank-r matrix $A \in \mathbb{R}^{m\times n}$ factors as $A = BC^T$ with $B \in \mathbb{R}^{m\times r}$ (a basis of the range) and $C \in \mathbb{R}^{n\times r}$ (coordinate matrix). Useful for approximating A when $r \ll m, n$.

16 Symmetric Matrices and Quadratic Forms

16.1 Symmetric Matrices

- A is **symmetric** if $A^T = A$ (necessarily square; entries mirror across the diagonal).
- **Real eigenvalues**: for $A = A^T$ and $Ax = \lambda x$ with $x \neq 0$, $\lambda = \dfrac{x^T Ax}{x^T x} \in \mathbb{R}$.
- Eigenvectors from *distinct* eigenspaces are orthogonal; for symmetric matrices geometric multiplicity = algebraic multiplicity.
- **Spectral theorem**: A is **orthogonally diagonalizable** ($A = Q\Lambda Q^T$, Q orthogonal) $\iff$ A is symmetric.
- For a symmetric matrix, the signs of the pivots match the signs of the eigenvalues (and both products equal $\det A$).

16.2 Quadratic Forms

- A **quadratic form** on $\mathbb{R}^n$ is $\mathcal{Q}(x) = x^T Ax$ with A symmetric:

$$x^T Ax = \sum_i \sum_j A_{ij} x_i x_j.$$

- **Reading off** A: diagonal entry A_{ii} is the coefficient of x_i^2; off-diagonal $A_{ij} = A_{ji}$ is *half* the coefficient of $x_i x_j$. Any B can be symmetrized without changing the form via $A = \frac{1}{2}(B + B^T)$.
- **Change of variable** $x = Py$ (P invertible): the form becomes $y^T(P^T AP)y$. Since A is symmetric, an orthogonal P makes $P^T AP = D$ diagonal, eliminating cross terms: $\mathcal{Q} = \sum_i \lambda_i y_i^2$.

16.3 Definiteness and Classification

A symmetric A (equivalently its form $\mathcal{Q}$) is:

- **Positive definite** ($A \succ 0$): $x^T Ax > 0$ for all $x \neq 0$ $\iff$ all eigenvalues > 0.
- **Positive semidefinite** ($A \succeq 0$): $x^T Ax \geq 0$ $\iff$ all eigenvalues ≥ 0.
- **Negative (semi)definite**: reverse the inequalities / eigenvalue signs.
- **Indefinite**: A has both positive and negative eigenvalues (the form takes both signs).

16.4 Positive-Definiteness Tests

For symmetric A, the following are equivalent statements that A is positive definite:

1. $x^T Ax > 0$ for all $x \neq \mathbf{0}$.
2. $A = S^T S$ for some S with linearly independent columns.
3. Every eigenvalue is > 0.
4. **Sylvester's criterion**: every leading principal minor (upper-left $k \times k$ determinant) is > 0.
5. Every pivot is > 0.

For positive *semi*definiteness, replace "> 0" by "≥ 0" and use *all* principal minors (not just leading ones).

16.5 Gram Matrices and Properties

- The **Gram matrix** $A^T A$ is symmetric with real, nonnegative eigenvalues and orthonormal eigenvectors — always positive semidefinite. It is positive definite $\iff$ the columns of A are linearly independent.
- If A, B are positive (semi)definite and $c > 0$: $A + B$, cA, A^T, and $P^* AP$ are positive (semi)definite. $P^* AP$ is positive *definite* iff $\operatorname{rank}(P) = m$.
- Positive/negative definite matrices are full rank, hence invertible; $A \succ 0 \Rightarrow A^{-1} \succ 0$.

17 Singular Value Decomposition (SVD)

17.1 Singular Values and Vectors

- For any $S \in \mathbb{R}^{m\times n}$, the **singular values** are $\sigma_i = \sqrt{\lambda_i}$ where λ_i are the (nonnegative) eigenvalues of S^TS, ordered $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.
- If $v_1, \ldots, v_n$ are orthonormal eigenvectors of S^TS, then $\|Sv_i\| = \sigma_i$.
- If $\sigma_1 \geq \cdots \geq \sigma_r > 0$ (with $\sigma_{r+1} = \cdots = 0$), then $\{Sv_1, \ldots, Sv_r\}$ is an orthogonal basis of range(S), so rank$(S) = r$ = number of nonzero singular values.
- Define $u_i = \dfrac{Sv_i}{\sigma_i}$ for $i \leq r$: the u_i are orthonormal ($Sv = \sigma u$), forming a basis of range(S).

17.2 The Decomposition

Every $A \in \mathbb{R}^{m\times n}$ factors as

$$A = U\Sigma V^T \qquad (\text{or } U\Sigma V^* \text{ over } \mathbb{C}),$$

where

- U is $m \times m$ orthogonal — columns (**left singular vectors**) are eigenvectors of AA^T;
- V is $n \times n$ orthogonal — columns (**right singular vectors**) are eigenvectors of A^TA;
- Σ is $m \times n$ diagonal with nonnegative singular values σ_i.

Consistency checks: $A^TA = V\Sigma^2V^T$ and $AA^T = U\Sigma^2U^T$, so $\sigma_i^2 = \lambda_i$. The SVD always exists (even for rectangular / rank-deficient A) and is essentially unique up to permutation / repeated singular values.

17.3 Computing an SVD

1. Find eigenpairs (λ_i, v_i) of A^TA; normalize the v_i into V (right singular vectors).
2. Form Σ from $\sigma_i = \sqrt{\lambda_i}$, ordered decreasingly.
3. Recover U from $AV = U\Sigma$, i.e. $U = AV\Sigma^{-1}$ (columns are the $u_i = Av_i/\sigma_i$, extended to an orthonormal basis of $\mathbb{R}^m$).

Reduced SVD: keeping only the $k = \min(m, n)$ meaningful directions gives $A = U_R\Sigma_RV_R^T$ with U_R $(m \times k)$, Σ_R $(k \times k)$, V_R $(n \times k)$; U_R, V_R have orthonormal columns but are no longer square.

17.4 Comparison with Other Factorizations

- **SVD** applies to *every* matrix (even rectangular), Σ is diagonal with real nonnegative entries, and it uses *two* orthogonal/unitary matrices U, V.
- **Diagonalization** needs a basis of eigenvectors and one invertible matrix.
- **Spectral decomposition** needs a normal matrix and one unitary matrix.

17.5 Consequences and Applications

- **Determinant**: $|\det A| = \prod_i \sigma_i$ (since $|\det U| = |\det V^*| = 1$).
- **Outer-product form**: $A = \sum_{i=1}^{r} \sigma_i u_i v_i^T$, and $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$ has rank$(A_k) = k$.
- **Best rank-k approximation**: keep the k largest singular values (zero the rest); this is the closest rank-k matrix to A (Eckart–Young).
- **Four fundamental subspaces**: $\{u_1, \ldots, u_r\}$ is an orthonormal basis of range(A); $\{u_{r+1}, \ldots, u_m\}$ of $N(A^*)$; $\{v_1, \ldots, v_r\}$ of range(A^*); $\{v_{r+1}, \ldots, v_n\}$ of $N(A)$.
- If A is positive semidefinite, its singular values equal its eigenvalues.

17.6 Pseudo-Inverse

- The **Moore–Penrose pseudo-inverse** is $A^+ = V\Sigma^+ U^T$, where Σ^+ inverts the nonzero singular values and leaves zeros in place: $(\Sigma^+)_i = 1/\sigma_i$ if $\sigma_i \neq 0$, else 0.
- It is defined for all matrices (non-square, singular, ...). When A^TA is invertible (tall, full rank), $A^+ = (A^TA)^{-1}A^T$ gives the least-squares solution; when AA^T is invertible (wide, full rank), $A^+ = A^T(AA^T)^{-1}$ gives the least-norm solution.
- In general $x = A^+b$ is the minimum-norm least-squares solution.

18 Vector and Matrix Norms

18.1 The p-Norm Family

- A **norm** $f : \mathbb{R}^n \to \mathbb{R}$ satisfies positivity ($f(x) \geq 0, = 0 \iff x = \mathbf{0}$), homogeneity ($f(\alpha x) = |\alpha| f(x)$), and the triangle inequality ($f(x+y) \leq f(x) + f(y)$).
- **p-norm** ($p \geq 1$): $\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p}$.
  - $\|x\|_1 = \sum_i |x_i|$ (Manhattan distance).
  - $\|x\|_2 = \sqrt{\sum_i x_i^2}$ (Euclidean).
  - $\|x\|_\infty = \max_i |x_i|$ (max norm).
- **Not true norms**: $\|x\|_0 = \#\{i : x_i \neq 0\}$ (counts nonzeros; violates homogeneity) and $\|x\|_{1/2}$ (fails the triangle inequality). The unit "ball" $\|x\|_p = 1$ is convex only for $p \geq 1$.

18.2 L^1 vs. L^2 and Sparsity

- L^1 regularization tends to produce **sparse** solutions (its diamond-shaped unit ball has corners on the axes), while L^2 spreads weight out and is more stable.
- Since $\|x\|_0$ minimization is NP-hard, the ℓ_1 norm is used as a **convex relaxation**: $\min_x \|x\|_1$ s.t. $\|Ax - b\|_2 < \epsilon$.

18.3 Norm Inequalities

For all $x \in \mathbb{R}^d$:

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{d}\,\|x\|_2, \qquad \|x\|_\infty \leq \|x\|_1 \leq d\,\|x\|_\infty, \qquad \|x\|_\infty \leq \|x\|_2 \leq \sqrt{d}\,\|x\|_\infty.$$

Every inner product induces a norm $\|x\| = \sqrt{\langle x, x\rangle}$, but not every norm comes from an inner product (e.g. $\|\cdot\|_\infty$).

18.4 Matrix Norms

- **Entrywise / Frobenius**: $\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2} = \sqrt{\text{trace}(A^*A)} = \left(\sum_i \sigma_i^2\right)^{1/2}$. It is invariant under unitary multiplication: $\|A\|_F = \|UA\|_F = \|AU\|_F$.
- **Induced (operator) norm**: $\|A\|_p = \max_{x \neq 0} \dfrac{\|Ax\|_p}{\|x\|_p}$, so $\|Ax\| \leq \|A\|\,\|x\|$ and $\|AB\| \leq \|A\|\,\|B\|$ (submultiplicative).
- Special cases: $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (max absolute column sum); $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (max absolute row sum).

19 Vector and Matrix Calculus

19.1 Fields, Gradients, and Jacobians

- A **scalar field** is $f : \mathbb{R}^n \to \mathbb{R}$; a **vector field** $f : \mathbb{R}^n \to \mathbb{R}^m$; a **matrix field** maps into matrices. Differentiating a scalar field gives the gradient (a vector); differentiating again gives the Hessian (a matrix).
- **Gradient / Jacobian** (numerator layout): for $f : \mathbb{R}^n \to \mathbb{R}$, the derivative $\dfrac{\partial f}{\partial x}$ is a *row* vector. Machine-learning texts often use the transposed "denominator layout"; the chain rule reverses order accordingly.
- **Directional derivative**: for $f : \mathbb{R}^n \to \mathbb{R}$ and direction v, $D_v f = v \cdot \nabla f$.
- **Jacobian**: for $f : \mathbb{R}^n \to \mathbb{R}^m$, $(J)_{ij} = \partial f_i / \partial x_j$; the **Hessian** of a scalar field is the Jacobian of its gradient.

19.2 Useful Identities

- Vector-by-vector (numerator layout):

$$\frac{\partial (Ax)}{\partial x} = A, \qquad \frac{\partial (x^T a)}{\partial x} = a^T, \qquad \frac{\partial (x^T A x)}{\partial x} = x^T (A + A^T) \ \big(= 2x^T A \text{ if } A = A^T \big).$$

- **Product rule** (with x, y functions of z): $\dfrac{\partial (x^T y)}{\partial z} = x^T \dfrac{\partial y}{\partial z} + y^T \dfrac{\partial x}{\partial z}$.
- Matrix-valued derivatives:

$$\frac{\partial \det(A)}{\partial A} = \det(A)\, A^{-1}, \qquad \frac{\partial \ln |\det A|}{\partial A} = A^{-1}, \qquad \frac{\partial (A^{-1})}{\partial t} = -A^{-1} \frac{\partial A}{\partial t} A^{-1}.$$

$$\frac{\partial (y^T A x)}{\partial A} = x y^T, \qquad \frac{\partial (x^T A x)}{\partial A} = x x^T.$$

20 Projections and Orthogonal Matrices

20.1 Projection onto a Vector / Subspace

- **Onto a vector**: for $u \neq \mathbf{0}$, $\operatorname{proj}_u(v) = \dfrac{u \cdot v}{u \cdot u}\, u$.
- **Orthogonal decomposition theorem**: if W has orthogonal basis $u_1, \ldots, u_p$, then every y splits uniquely as $y = \hat{y} + z$ with $\hat{y} \in W$, $z \in W^{\perp}$, where

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \cdots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p$$

  is the orthogonal projection of y onto W.

20.2 Orthogonal Matrices

- An **orthogonal matrix** has orthonormal columns. If *square*: $A^T A = A A^T = I$ and $A^{-1} = A^T$.
- **Tall** $A \in \mathbb{R}^{m \times n}$, $m > n$: $A^T A = I_n$ but $A A^T \neq I_m$ ($A A^T$ is the projection onto $C(A)$).
- **Wide** A, $m < n$: $A A^T = I_m$ but $A^T A \neq I_n$.
- The map $f(x) = Ax$ (orthonormal columns) preserves inner products, norms, distances, and angles: $(Ax)^T (Ay) = x^T y$.
- Every 2×2 orthogonal matrix is a rotation $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ or a reflection $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$.

20.3 Projection Matrices

- For A with independent columns, the orthogonal projection onto $C(A)$ is
$$P = A(A^TA)^{-1}A^T.$$
- P is **symmetric** ($P^T = P$) and **idempotent** ($P^2 = P$); it fixes vectors in $C(A)$ and annihilates vectors in $C(A)^\perp = N(A^T)$.
- This rests on the fundamental orthogonality $C(A) \perp N(A^T)$ and $R(A) \perp N(A)$.

21 Least Squares

21.1 The Problem

When $Ax = b$ has no solution ($b \notin C(A)$), we instead minimize the residual $\|Ax - b\|$. The bad news is $Ax = b$ is unsolvable; the good news is $A\hat{x} = \hat{b}$ (with $\hat{b}$ the projection of b onto $C(A)$) always is.

21.2 Normal Equations (Method 1)

- The residual $b - A\hat{x}$ must be orthogonal to $C(A)$, i.e. lie in $N(A^T)$: $A^T(b - A\hat{x}) = \mathbf{0}$. This gives the **normal equations**
$$A^TA\,\hat{x} = A^Tb.$$
- If A has linearly independent columns then A^TA is invertible and
$$\hat{x} = (A^TA)^{-1}A^Tb = A^+b,$$
the unique least-squares solution; $(A^TA)^{-1}A^T$ is the left inverse of A, and $A(A^TA)^{-1}A^T$ is the projection onto $C(A)$.
- When A is square and invertible this reduces to $\hat{x} = A^{-1}b$.

21.3 Derivation and QR (Methods 2–3)

- **Calculus**: minimizing $f(x) = (Ax-b)^T(Ax-b)$ via $\nabla f(x) = 0$ gives $2A^TAx - 2A^Tb = 0$, i.e. the same normal equations (using $\partial(x^TBx)/\partial x = 2x^TB$ for symmetric B).
- **QR**: with $A = QR$, solve $Rx_{LS} = Q^Tb$ by back substitution. Cost $\approx 2mn^2$; numerically more stable than forming A^TA.

21.4 Rank-Deficient Case and SVD (Method 4)

- A^TA is invertible $\iff$ A has linearly independent columns (typical when the number of equations N greatly exceeds the number of unknowns D).
- When A^TA is singular (e.g. $N < D$), the three methods above fail and there are infinitely many solutions. Using the SVD $A = U\Sigma V^T$,
$$x = V\Sigma^+U^Tb$$
is the **minimum-norm** least-squares solution; the full solution set adds any element of $N(A)$.

21.5 Weighted and Constrained Least Squares

- **Weighted**: minimizing $\sum_k \|\lambda_k(A_kx - b_k)\|^2$ stacks the weighted blocks; if the stacked matrix has independent columns,
$$\hat{x} = \Big(\textstyle\sum_k \lambda_k A_k^TA_k\Big)^{-1}\Big(\sum_k \lambda_k A_k^Tb_k\Big).$$
- **Constrained** ($\min \|Ax - b\|^2$ s.t. $Cx = d$): form the Lagrangian $L(x, \lambda) = \|Ax - b\|^2 + \lambda^T(Cx - d)$ and solve the KKT system
$$\begin{bmatrix} 2A^TA & C^T \\ C & 0 \end{bmatrix}\begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} 2A^Tb \\ d \end{bmatrix}.$$

22 Cheat Sheet

Core objects

- Ax = lin. comb. of columns of A; $(Ax)_i = \tilde{a}_i^T x$.
- $C(A)$ = span of columns; $R(A) = C(A^T)$; $N(A) = \ker A$; $N(A^T)$ = left null space.
- $\operatorname{rank}(A) = \dim C(A) = \dim R(A) = r$.
- **Rank–nullity**: $\operatorname{rank}(A) + \dim N(A) = n$.
- **Four subspaces**: $C(A), N(A^T) \subseteq \mathbb{R}^m$; $R(A), N(A) \subseteq \mathbb{R}^n$; $C(A) \perp N(A^T)$, $R(A) \perp N(A)$.

Independence / basis

- LD $\iff$ some vector is a lin. comb. of the others; contains $\mathbf{0} \Rightarrow$ LD.
- Coeffs in a basis are unique; two bases have equal size $= \dim V$.
- $p > \dim V$ vectors $\Rightarrow$ LD; $p < \dim V$ can't span.
- $\dim(W_1{+}W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.
- $\dim F^n{=}n$, $\dim P_p{=}p{+}1$, $\dim M_{m,n}{=}mn$.

Solving $Ax = b$

- Consistent $\iff$ $b \in C(A)$ $\iff$ last column not a pivot in $[A \mid b]$.
- General solution = particular + $N(A)$.
- Homogeneous $Ax = \mathbf{0}$ has nontrivial solutions $\iff \operatorname{rank}(A) < n$.
- $N(A) = N(A^TA)$, so $\operatorname{rank}(A^TA) = \operatorname{rank}(A)$; $\operatorname{rank}(A) = \operatorname{rank}(A^T)$.

Invertible Matrix Thm ($A_{n\times n}$)

Equivalent: invertible $\iff$ $\det A \neq 0$ $\iff$ $\operatorname{rank} = n$ $\iff$ RREF $= I$ $\iff$ cols (rows) independent & span $\iff$ $Ax{=}\mathbf{0}$ only trivial $\iff$ $x \mapsto Ax$ 1-1 & onto $\iff$ $0 \notin \sigma(A)$ $\iff$ has left & right inverse.

Linear maps

- $\dim V = \operatorname{nullity}(T) + \dim \operatorname{range}(T)$.
- 1-1 $\iff$ $N(T) = \{\mathbf{0}\}$ $\iff$ cols indep; onto $\iff$ cols span.
- Square: 1-1 $\iff$ onto. Wide: never 1-1; tall: never onto.
- $[T]_B = P^{-1}[T]_E P$ (change of basis); $R_\theta^n = R_{n\theta}$.

Inverse

- A^{-1} exists $\iff$ $\det A \neq 0$ $\iff$ $\operatorname{rank} = n$ $\iff$ RREF $= I$.
- $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$.
- $A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$, $\operatorname{adj}(A) = C^T$.

Determinant

- $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.
- Cofactor: $\det A = \sum_j a_{ij}C_{ij}$, $C_{ij} = (-1)^{i+j}\det A_{ij}$.
- $\det(AB) = \det A \det B$; $\det(A^T) = \det A$; $\det(kA) = k^n \det A$; $\det(A^{-1}) = 1/\det A$.
- Triangular $\Rightarrow$ det $=$ $\prod$ diagonal; swap flips sign; add-multiple keeps it.
- Cramer: $x_i = \det A_i(b)/\det A$.

Eigenvalues

- $Av = \lambda v$, $v \neq 0$; $E_\lambda = N(A - \lambda I)$; solve $\det(A - \lambda I) = 0$.
- $\sum \lambda_i = \operatorname{trace}(A)$; $\prod \lambda_i = \det A$.
- Triangular: λ_i = diagonal. $0 \in \sigma(A)$ $\iff$ singular.
- geo. mult. $\leq$ alg. mult.

Diagonalization

- $A = Q\Lambda Q^{-1}$ $\iff$ n indep. eigenvectors; n distinct $\lambda \Rightarrow$ yes.
- $A^k = Q\Lambda^k Q^{-1}$.
- Similar $A = Q^{-1}BQ$: same $\det, \operatorname{tr}, \operatorname{rank}$, char. poly, λ.
- Cayley–Hamilton: $p(A) = 0$.

Symmetric / quadratic

- $A = A^T \Rightarrow$ real λ, orthonormal eigenvectors, $A = Q\Lambda Q^T$.
- $\mathcal{Q}(x) = x^TAx$; diag. entry = coeff. x_i^2, off-diag. $= \frac{1}{2}$ cross term.
- PD $\iff$ all $\lambda > 0$ $\iff$ leading minors > 0 $\iff$ pivots > 0 $\iff$ $A = S^TS$ (indep. cols).
- $A^TA \succeq 0$ always; $\succ 0$ $\iff$ cols independent.

Orthogonality

- Orthogonal set of nonzero vectors is independent.
- Orthonormal expansion: $y = \sum_i (a_i^T y)a_i$; orthogonal: $c_j = \frac{y \cdot a_j}{a_j \cdot a_j}$.
- $\operatorname{proj}_u v = \frac{u \cdot v}{u \cdot u}u$; projection onto $C(A)$: $P = A(A^TA)^{-1}A^T$, $P^2 = P = P^T$.
- Orthogonal A: $A^TA = I$, preserves norm/angle; square $\Rightarrow A^{-1} = A^T$.
- Gram–Schmidt: $\tilde{q}_i = a_i - \sum_{j<i}\langle q_j, a_i\rangle q_j$, $q_i = \tilde{q}_i/\|\tilde{q}_i\|$.

Factorizations

- $A = QR$: $Q^TQ = I$, R upper-$\triangle$, $R = Q^TA$.
- $A = LU$ (leading minors $\neq 0$); $A = PLU$ general.
- $A = R^TR$ (Cholesky, PD).
- Normal $\Rightarrow A = UDU^*$; symmetric $\Rightarrow A = UDU^T$; general $\Rightarrow$ SVD.

SVD

- $A = U\Sigma V^T$; $\sigma_i = \sqrt{\lambda_i(A^TA)}$; V: eigvecs of A^TA, U: eigvecs of AA^T, $u_i = Av_i/\sigma_i$.
- rank $= \#\{\sigma_i > 0\}$; $|\det A| = \prod \sigma_i$; $\|A\|_F^2 = \sum \sigma_i^2$.
- $A = \sum_i \sigma_i u_i v_i^T$; best rank-$k$: keep top k σ_i.
- $A^+ = V\Sigma^+U^T$ ($1/\sigma_i$ if $\sigma_i \neq 0$, else 0).

Norms

- $\|x\|_p = (\sum |x_i|^p)^{1/p}$; $\|x\|_1, \|x\|_2, \|x\|_\infty = \max|x_i|$.
- $\|x\|_2 \leq \|x\|_1 \leq \sqrt{d}\|x\|_2$; $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{d}\|x\|_\infty$.
- $\|A\|_F = \sqrt{\operatorname{tr}(A^*A)}$; $\|A\|_1 =$ max col sum, $\|A\|_\infty =$ max row sum; $\|AB\| \leq \|A\|\|B\|$.

Calculus

- $\partial(Ax)/\partial x = A$; $\partial(x^Ta)/\partial x = a^T$.
- $\partial(x^TAx)/\partial x = x^T(A + A^T) = 2x^TA$ (sym).
- $\partial \det A/\partial A = \det A\, A^{-1}$; $\partial \ln|\det A|/\partial A = A^{-1}$.

Least squares

- Normal eq.: $A^TA\hat{x} = A^Tb \Rightarrow \hat{x} = (A^TA)^{-1}A^Tb = A^+b$.
- Residual $b - A\hat{x} \perp C(A)$ (in $N(A^T)$).
- Rank-deficient: $x = V\Sigma^+U^Tb$ (min-norm).